

Derivative of sigmoid function:

$$G(x) = \frac{1}{1+e^{-x}}$$

$$\Rightarrow G(x) = (1+e^{-x})^{-1}$$

$\therefore$ According to chain rule:

$$\frac{d}{dx} f(g(x)) = f'(g(x)) \cdot g'(x)$$

Suppose,

$$u = \cancel{e^{-x}} 1 + e^{-x}$$

$$\therefore f(u) = u^{-1}$$

$$\therefore \frac{d}{dx} f(u) = -1 u^{-2} = -(1+e^{-x})^{-2}$$

$$\begin{aligned} \text{Again, } \frac{d}{dx} u &= \frac{d}{dx} (1+e^{-x}) \\ &= -e^{-x} \end{aligned}$$

$$\begin{aligned} \therefore \text{Applying chain rule: } & \begin{array}{cc} \text{outer} & \text{inner} \\ \uparrow & \uparrow \end{array} \\ \frac{d}{dx} G(x) &= -(1+e^{-x})^{-2} \cdot (-e^{-x}) \\ &= \frac{e^{-x}}{(1+e^{-x})} \end{aligned}$$

∴ Converting into sigmoid form:

$$G(x) = \frac{1}{1+e^{-x}} \quad \text{and} \quad 1-G(x) = \frac{e^{-x}}{1+e^{-x}}$$

$$\begin{aligned} \therefore G(x) \cdot (1-G(x)) &= \frac{1}{(1+e^{-x})} \cdot \frac{e^{-x}}{(1+e^{-x})} \\ &= \frac{e^{-x}}{(1+e^{-x})^2} \end{aligned}$$

$$\therefore G'(x) = G(x)(1-G(x))$$

or

$$G'(x) = \text{output} (1 - \text{output})$$